Draft

2. Literature review

Rainfall generation under future climate condition require models that can capture the full range of rainfall variability, including sub-daily fluctuations, seasonal patterns, and extremes. This capability is critical for applications such as urban drainage design, flood forecasting, water resources planning, and climate risk assessment (Fatichi, Ivanov & Caporali, 2011; Onof, C. et al., 2000). However, observational rainfall records are often short, incomplete, or limited in spatial coverage, especially at fine temporal resolutions. These limitations motivate the use of stochastic rainfall generators, which can simulate synthetic time series that statistically resemble observed rainfall (Kim et al., 2017; Wilks, 1998). Unlike climate model outputs, stochastic rainfall generators can create large ensembles of synthetic sequences at sub-hourly resolution. This capability enables probabilistic assessment of future rainfall behaviour, which is essential for robust infrastructure planning and uncertainty analysis under climate change. Various stochastic rainfall models have been proposed, including Neyman-Scott rectangular pulse models, Markov chain-based models, and two-part weather generators (Cowpertwait, PSP et al., 1996). Among these, Poisson cluster-based models offer a compelling combination of physical interpretability and statistical performance. A particularly prominent subclass is the Bartlett-Lewis Rectangular Pulse (BLRP) model, originally developed by RODRIGUEZ-ITURBE, COX & ISHAM, (1987), which has since undergone numerous refinements to improve its ability to simulate rainfall across temporal scale (Kim & Onof, 2020).

The original BLRP model, developed by RODRIGUEZ-ITURBE, COX & ISHAM, (1987), represents rainfall a series of clustered events generated by Poisson. Storms arrive according to a Poisson process, within which rain cells are also governed by Poisson arrivals, with fixed-duration rectangular pulses assigned random intensities. This structure is computationally efficient and captures important features of rainfall such as intermittency and clustering. However, the original formulation is limited in reproducing rainfall extremes and dry spell duration, especially at sub-daily timescales

(Onof, C. et al., 2000; RODRIGUEZ-ITURBE, COX & ISHAM, 1988).

Over the past 30 years, significant refinements to the BLRP model have been developed to address these limitations. One of the key developments was the Randomized Bartlett-Lewis Rectangular Pulse model (RBL1), which introduced randomness in the rain cell duration parameter η. This adjustment improved the model's ability to represent dry and wet spell statistics across various temporal resolutions (RODRIGUEZ-ITURBE, COX & ISHAM, 1988). Subsequent models have focused on improving the statistical reproduction of extremes and extending temporal resolution without increasing model complexity. Kaczmarska, Isham & Onof, (2014) proposed RBL2, where rainfall cell intensity is modelled as inversely proportional to cell duration, following physical observations that intense convective storms tend to be short-lived. This refinement significantly improved performance at sub-hourly timescales without altering the rectangular pulse structure.

With increasing emphasis on climate change adaptation, rainfall generators must reflect long-term climate shifts while preserving short-term rainfall variability. Further extensions have attempted to improve the model's performance over longer timescales. Park, Onof & Kim, (2019) introduced a hybrid model (RBL3) by combining RBL1 for hourly rainfall generation with a Seasonal ARIMA (SARIMA) model for monthly rainfall, capturing memory and variability form hours to years. (Onof, Christian & Wang, 2020) developed RBL4, improving the estimation of standard rainfall statistics by deriving new analytical expressions and more efficient calibration methods. One of the most important developments came with RBL5 by Kim & Onof, (2020), which introduced a storm 'shuffling' algorithm to incorporate correlation between consecutive storm depth, enabling the reproduction of rainfall variability from 5-minute to decadal scales. Their three-model structure retained intra-storm cell dynamics, sub-monthly sequencing, and monthly-scale rainfall properties, overcoming the tendency of Poisson models to underestimate coarse-scale variance and skewness.

Recognizing the limitations of rectangular pulses in representing rainfall intensity dynamics, Park et al. (2021) proposed RBL6 by replacing the rectangular pulses with a sine-square ($\sin^2$) pulse. This modification allowed for more realistic intensity variation within rain cells and yielded improved performance in reproducing rainfall statistics, particularly extremes, at sub-hourly resolutions. Building on this idea, Vuong Tai et al. (preprint) introduced the RBL7 model, which uses a variable sine-k pulse where the shape of the pulse changes with rain cell duration. The shorter the duration, the more peaked the pulse become, reflecting the nature of convective events. This dynamic representation allows the model to maintain accuracy in both typical and extreme rainfall conditions using only hourly and supra-hourly calibration data, which is particularly valuable in data- scarce contexts.

These successive modifications reflect the evolution of the Bartlett-Lewis model from a simple Poisson cluster structure to a flexible and multi-scale rainfall generator. The key structural innovation and performance improvements of each Bartlett-Lewis model variant are summarised in Table 1 below.

Model	**Key Feature/ Structure Innovation**	**Main Improvement/ capability**	**Reference**
BLRP (OBL)	Storms and rain cells follow Poisson processes; constant duration rectangular pulse	Captures basic rainfall intermittency and clustering	(RODRIGUEZ-ITURBE, COX & ISHAM, 1987)
RBL1	Introduces randomness to ran cell duration (η)	Improve simulation of dry/ wet spell durations across timescales	(RODRIGUEZ-ITURBE, COX & ISHAM, 1988)
RBL2	Rainfall intensity inversely population to cell duration	Reflects convective storm physics; enhance sub-hourly performance	(Kaczmarska, Isham & Onof, 2014)

RBL3	Combine RBL1 with SARIMA model at monthly scale	Capture seasonal variability and memory from hours to year	(Park, Onof & Kim, 2019)
RBL4	New analytical expressions and improve fitting procedure	Enhance calibration accuracy for standard statistics (e.g. variance, skewness)	(Onof, Christian & Wang, 2020)
RBL5	Adds storm-depth huffling across events	Preserves storm-to-storm correlation; effective from 5-min to decadal scales	(Kim & Onof, 2020)
RBL6	Replaces rectangular pulse with sine-square ($\sin^2$) pulse	Allows variable intensity within rain cell; better extreme rainfall simulation	(Park et al., 2021)
RBL7	Variable sin-k pulse shape depending on cell duration	Reflects peakedness of convective storms; high accuracy using limited data	(Vuong Tai et al.)

A common challenge across all BLRP-type model is parameter calibration. The parameter space (usually consisting of 6 to 8 variables) is high-dimensional and often contains multiple local optima. Early approaches relied on genetic algorithms or manual trial and error, but these proved inefficient or inconsistent (Cowpertwait, PSP et al., 1996). Recently, optimization schemes such as Particles Swarm Optimization (PSO) and hybrid techniques combing PSO with local gradient-based methods have been used to improve calibration efficiency and robustness (Kim et al., 2017; Vuong Tai et al., preprint). In particular, weighting rainfall statistics inversely proportional to their inter-annual variance, as proposed by Kaczmarska, Isham & Onof, (2014), has become a standard approach to balance different statistical targets (e.g. mean, variance, skewness, dry period frequency) during calibration.

The UK Climate Projections 2018 (UKCP18) include a high-resolution Convection-Permitting Model (CPM) ensemble that provides hourly rainfall data at a 2.2 km spatial resolution (Met Office, 2019). The CPM is produced by dynamically downscaling a Regional Climate Model (RCM), which itself is driven by boundary conditions from a Global Climate Model (GCM), using the Met Office Unified Model. This nested framework enables the explicit simulation of convective processes without relying on parameterisation schemes (Kendon, Elizabeth J. et al., 2020). Unlike conventional RCMs, the UKCP18 CPM resolves convective processes explicitly, improving the representation of sub-daily rainfall features, such as intense summer storms and short-duration extremes (Kendon, Elizabeth J. et al., 2020; Kennedy-Asser et al., 2021). This results in more realistic simulations of storm cell dynamics, sub-hourly variability, and localized intensity peaks—features often missed or smoothed in coarser models (Chan et al., 2016). These capabilities make CPMs particularly valuable tools for high-resolution flood risk assessment and urban drainage system design. However, the direct use of CPM outputs in stochastic rainfall generation remains limited. High computational demands constrain ensemble size and simulation length, with most runs restricted to short time slices (e.g. 1981–2000 or 2021–2040) (Chan et al., 2016). Moreover, each ensemble member is a deterministic realisation, lacking the probabilistic variability necessary for scenario analysis and risk-based planning. This limitation, along with residual systematic biases—particularly in extremes and dry periods—necessitates the application of post-processing techniques. Statistical approaches such as quantile mapping, empirical cumulative distribution adjustment, and the widely used Change Factor (CF) method are commonly applied to align climate model outputs with observed rainfall statistics. These methods allow the integration of climate projections into stochastic modelling frameworks that require stationarity, high temporal resolution, and ensemble flexibility. As a result, coupling CPM outputs with stochastic generators remains essential for generating plausible future rainfall scenarios under climate change.

To bridge this scale gap, statistical downscaling and bias correction techniques such as the Change Factor (CF) approach are often applied, whereby projected changes in rainfall statistics (e.g. mean, variance) from climate models are applied to historical observation. These modified series can then be used as inputs or calibration targets for stochastic rainfall generators (Fatichi, Ivanov & Caporali, 2011; Wilks, 1998). Hybrid approaches have become common: for example, Fatichi, Ivanov & Caporali, (2011) coupled autoregressive models with Poisson cluster generators to reproduce interannual and sub-daily variability, while Park, Onof & Kim, (2019) applied a SARIMA model to monthly-scale rainfall linked to climate predictors, then used RBL-type generators to simulate hourly sequences. Similarly, Kim & Onof, (2020) developed a modular framework in which seasonal rainfall behavior is handled separately from fine-scale variability using a shuffling-enhanced Bartlett-Lewis model. Such multiscale structures allow integration of climate change signals into stochastic rainfall generators while maintaining statistical fidelity and flexibility.

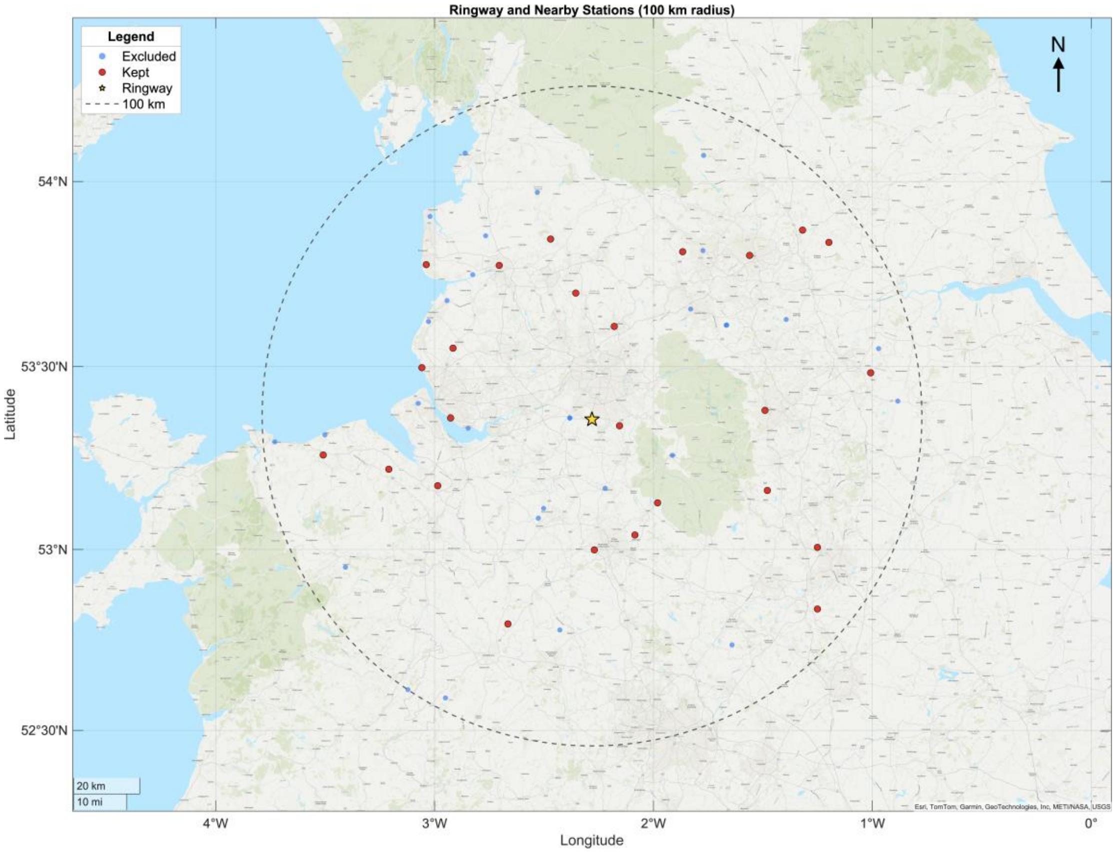
In summary, the Bartlett–Lewis model family has evolved into a robust and versatile framework for stochastic rainfall generation across temporal scales and climate conditions. From the original OBL formulation to recent advancements like RBL7, successive refinements have progressively improved the statistical realism, temporal resolution, and adaptability of these models. When coupled with climate projection techniques—particularly through modular and multi-scale structures-BLRP-type models offer a powerful means of generating synthetic rainfall sequences representative of future climate scenarios. These approaches bridge the gap between limited observational records, coarse-resolution climate models, and the fine-scale variability required for hydrological and infrastructure applications. Nonetheless, future research should focus on improving model calibration efficiency, integrating dynamic climate signals more systematically, and enhancing the flexibility of stochastic frameworks to accommodate uncertainty and extremes. Continued development in this direction is critical to supporting resilient water resource planning and flood risk management in a changing climate.

3. Study area and data

3.1 Study area and data

The primary site in this study is the Ringway meteorological station (SRC_ID: 1135), located in Greater Manchester, north-west England (53.353° N, 2.274° W; elevation ~69 m), situated within the premises of Manchester Airport, historically known as Ringway Airport. The station forms part of the UK Met Office's national meteorological network and has a long-term record of continuous hourly rainfall observations. The region experiences a temperate maritime climate, with frequent frontal systems driven by prevailing westerlies and occasional high-intensity convective storms during summer. Orographic enhancement from the nearby Pennines influences both seasonal totals and the structure of short-duration rainfall events (Burt & Ferranti, 2012).

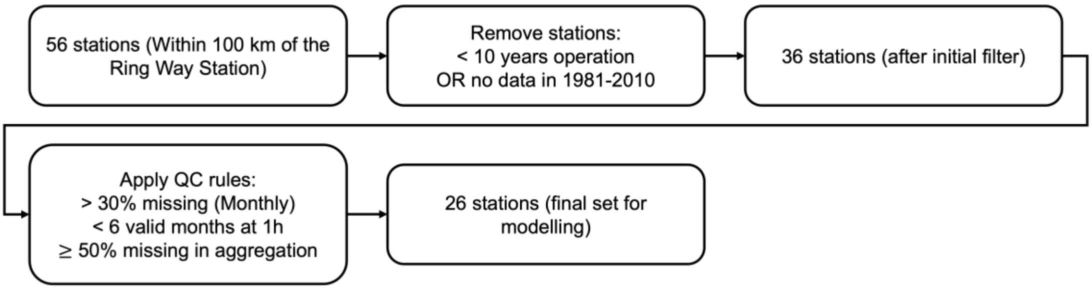
To improve the stability of statistical relationships used in the modelling framework, a set of surrounding stations within approximately 100 km of Ringway was also included. In total, 56 stations (including Ringway) were initially identified from the UK Met Office MIDAS hourly rainfall dataset. These gauges span a mixture of lowland and upland locations, providing a diverse representation of rainfall regimes in the region. The spatial distribution of all candidate stations is shown in Figure 1, with the final set of stations retained for modelling highlighted.



3.2 Rainfall observations and quality control

Hourly rainfall data for the period 1981–2010 were obtained from the UK Met Office MIDAS database (Met Office, 2012). This period matches the "historical" baseline used in UKCP18 climate projections and provides a consistent basis for comparing observed and simulated statistics. The original hourly data were processed in MATLAB to ensure consistent time formatting and to remove extraneous variables. NaN values were left as missing rather than replaced with zeros, to preserve the natural intermittency structure of rainfall (Onof, C. et al., 2000).

A stepwise quality control (QC) process was applied (Figure 2). First, stations with less than ten years of operation, or with no data during the 1981–2010 period, were removed. This reduced the set from 56 to 36 stations. Next, monthly subsets with more than 30 % missing hourly values were excluded. At the 1-hour scale, a month was considered valid if its mean rainfall exceeded 0.01 mm h^{-1}; stations with fewer than six valid months over the full 30-year period were removed. For aggregated durations (3 h, 6 h, 12 h, and 24 h), any period with ≥50 % missing constituent hours was discarded.



Following these steps, 26 stations, including Ringway, were retained for analysis. The combination of lenient completeness criteria and the use of a regional station pool was intended to maximize the number of data points for regression fitting, which is particularly beneficial when estimating higher-order rainfall statistics (Burton et al., 2010; Fowler et al., 2005). The locations, basic metadata, and mean annual rainfall for these stations are summarized in Table 1, which includes station ID, station name, latitude, longitude, elevation, and observation years.

Src id	Station name	Latitude (°N)	Longitude (°E)	Elevation (m)	Observation years
513	BINGLEY, NO 2	53.811	-1.867	262	31
523	LEEDS WEATHER CENTRE	53.801	-1.561	64	15
525	SHEFFIELD	53.381	-1.491	131	17
533	CHURCH FENTON	53.836	-1.199	8	32
534	BRAMHAM	53.869	-1.319	54	24
549	ASHOVER NO 2	53.162	-1.48	178	25
554	SUTTON BONINGTON	52.836	-1.251	43	25
556	NOTTINGHAM, WATNALL	53.006	-1.251	117	68
562	FINNINGLEY	53.483	-1.008	10	38
622	KEELE	52.999	-2.27	179	45
627	CELLARHEAD	53.04	-2.085	228	15
643	SHAWBURY	52.795	-2.665	72	68
1090	BLACKPOOL, SQUIRES GATE	53.776	-3.038	10	68
1093	AIGBURTH (RTC)	53.36	-2.927	12	13
1096	AUGHTON	53.55	-2.916	54	19
1111	PRESTON, MOOR PARK	53.774	-2.705	33	45
1119	STONYHURST	53.845	-2.47	115	25
1125	ROCHDALE	53.609	-2.179	110	18
1135	RINGWAY	53.356	-2.281	69	56
1137	RHYL NO 2	53.259	-3.509	77	33
1142	MOEL-Y-CRIO	53.22	-3.209	263	25

1144	HAWARDEN AIRPORT	53.175	-2.986	11	33
12001	HOLDEN WOOD RESR NO 2	53.699	-2.355	212	10
17309	CROSBY	53.497	-3.058	9	42
30690	LEEK, THORNCLIFFE	53.128	-1.981	298	23
55511	WOODFORD	53.339	-2.155	88	10

2.3 Climate projection data

Future climate information was sourced from the UKCP18 Convection-Permitting Model (CPM) ensemble (Kendon, E. J. et al., 2021). The CPM provides hourly precipitation at ~2.2 km resolution for both historical (1981–2000) and future (2061–2080) periods. All simulations are driven by the high-emissions Representative Concentration Pathway RCP8.5, which is consistent in radiative forcing with the CMIP6 SSP5–8.5 scenario. In the CMIP6 framework, SSP5-8.5 represents a fossil-fuel intensive development pathway with a radiative forcing of approximately 8.5 W m^{-2} by 2100, making it closely comparable to RCP8.5 used in UKCP18.

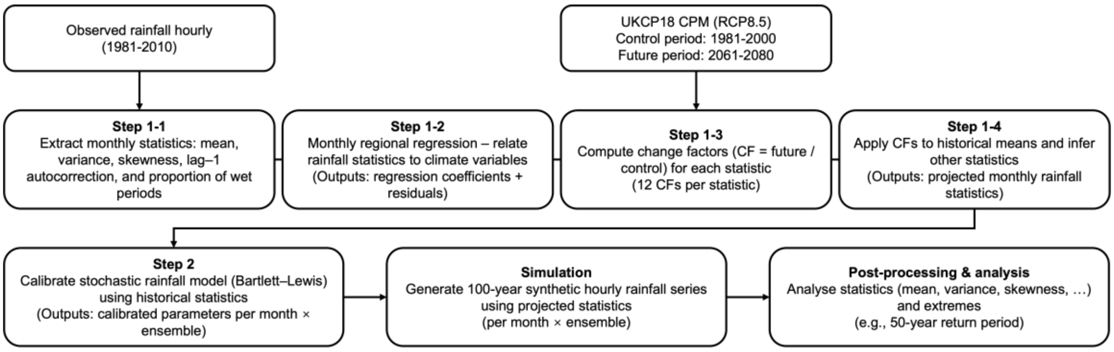
The CPM datasets were obtained from the UK Met Office UKCP18 archive via the Centre for Environmental Data Analysis (CEDA; https://data.ceda.ac.uk). For each station location, hourly precipitation was extracted from the central grid cell of the 3 × 3 CPM grid centred on the station coordinates, ensuring consistency between observed and projected series. In this study, only the first CPM ensemble member was used, following computational efficiency considerations and supervisory guidance. Details of how CPM outputs are processed and linked to observed statistics are provided in next section.

4. Methodology

4.1 Overview of modelling framework

Rainfall generation under a future climate scenario requires a modelling framework that can translate coarse-scale climate projections into long, high-resolution rainfall sequences. At the same time, it should retain the key statistical properties of the observed record. In this study, this transformation is achieved by combining long-term change information from the UKCP18 Convection-Permitting Model (CPM) with locally observed rainfall characteristics. Projected changes are first quantified. They are then applied to statistical descriptors of hourly rainfall so that the future sequences reflect both the large-scale climate signal and the fine-scale structure of the historical record.

The methodology is organized into a sequence of steps (Figure 3.1). These steps progressively transform historical observations and climate projections into long-term, high-resolution rainfall series for statistical and extreme value analysis.



Step 1 derives the target rainfall statistics for the future period: (1-1) Key monthly rainfall statistics are extracted from 1981–2010 hourly observations. These include mean, variance, skewness, lag-1 autocorrelation, and wet probability. Together, they capture the central tendency, spread, asymmetry, persistence, and intermittency of rainfall. (1-2) Linear regression relationships are established between mean rainfall and other statistics, and between mean rainfall and selected climate variables. Data from multiple stations in the study region are used to improve robustness and reduce bias. (1-3) Change factors (CFs) are calculated from CPM outputs to represent projected differences between the control period (1981–2000) and the future period (2061–2080). (1-4) CFs are applied to historical mean rainfall. The regression relationships are then used to estimate other future statistics.

Step 2 uses these projected statistics to calibrate a stochastic rainfall model. The model is run separately for each month and ensemble realization to generate 100-year synthetic hourly rainfall series. Calibration aims to reproduce the projected statistics as closely as possible. This ensures the generated series are statistically consistent with future projections and temporally realistic at the hourly scale. Post-processing of the generated sequences allows for detailed assessment of changes in rainfall characteristics, including extreme events such as 50-year return period rainfall.

4.2 Generation of future rainfall statistics

4.2.1 Extraction of historical rainfall statistics

The first step was to derive a set of rainfall statistics from observations. Hourly records from Ringway and nearby stations were used for the period 1981–2010. This 30-year baseline captures both seasonal cycles and interannual variability. Using multiple stations improves the stability of the estimates, especially for higher-order moments that are sensitive to sampling variability (Burton et al., 2010; Fowler et al., 2005). Although the CPM control period is 1981–2000, the longer observational baseline was retained to maximize data availability. Consistency between baselines is handled later when applying change factors.

Data quality control excluded periods with excessive missing data and removed outliers inconsistent with regional climatology, while genuine extremes were preserved. Missing values were left as NaN rather than replaced by 0, preserving the natural intermittency of rainfall (Onof, C. et al., 2000). Coarser time series (3 h, 6 h, 12 h, and 24 h) were created by summing consecutive hourly totals. For example, a 6 h rainfall amount is the sum of six successive hourly values. This approach makes it possible to analyse statistics across multiple temporal scales using a single underlying dataset.

Rainfall statistics were extracted at five durations: 1, 3, 6, 12 and 24 h. These durations are widely used in hydrological design and frequency analysis, as they correspond to critical accumulation periods for sewer surcharge, surface flooding, and river catchment response (Burton et al., 2010; Fowler et al., 2005). From a modelling perspective, Bartlett–Lewis type generators are typically calibrated against these durations to ensure multi-scale consistency (Kim et al., 2017).

Rainfall statistics were computes separately for each calendar month and for five accumulation durations: 1h, 3h, 6h, 12h, and 24h. for a given duration d, let $X_t^{(d)}$

demote the accumulated rainfall at time t, and let R_t be the hourly rainfall. The following statistics were derived:

$$\mu_d = E\left[X_t^{(d)}\right] \tag{1}$$

$$\sigma_d^2 = Var\left(X_t^{(d)}\right) \tag{2}$$

$$\gamma_d = \frac{E\left[(X_t^{(d)} - \mu_d)^3\right]}{(\sigma_d)^3} \tag{3}$$

$$c_d^{(1)} = Cov\left(X_t^{(d)}, X_{t-1}^{(d)}\right) \tag{4}$$

$$PW = \Pr(R_t > 0) = \frac{wet\ hours\ in\ the\ month}{valid\ hours\ in\ the\ month} \tag{5}$$

The wet probability (PW) therefore represents the proportion of hours in a month with non-zero rainfall, expressed either probabilistically or as a simple frequency ratio.

Each statistic has a different physical meaning. Mean rainfall reflects the average wetness of a period, while variance indicates the variations around the mean. Skewness highlights the asymmetry of the distribution, influenced by intense but rare storms. Lag-1 autocovariance measures short term persistence, associated with storm clustering. Finally, wet probability (PW) quantifies intermittency by describing the frequency of rainfall occurrence. These statistics form the calibration targets for Poisson-cluster models such as the Bartlett–Lewis Rectangular Pulse family (Kim et al., 2017; RODRIGUEZ-ITURBE, COX & ISHAM, 1987).

To illustrate the extraction of historical statistics, Figure X presents the monthly mean rainfall at Ringway station together with the regional mean of 26 surrounding stations for the period 1981–2010. Both series exhibit a clear seasonal cycle, with higher rainfall during the winter months and lower totals in summer. The close agreement between Ringway and the regional average highlights that the selected reference site is representative of the wider area, providing a robust baseline for the subsequent regression and projection steps.

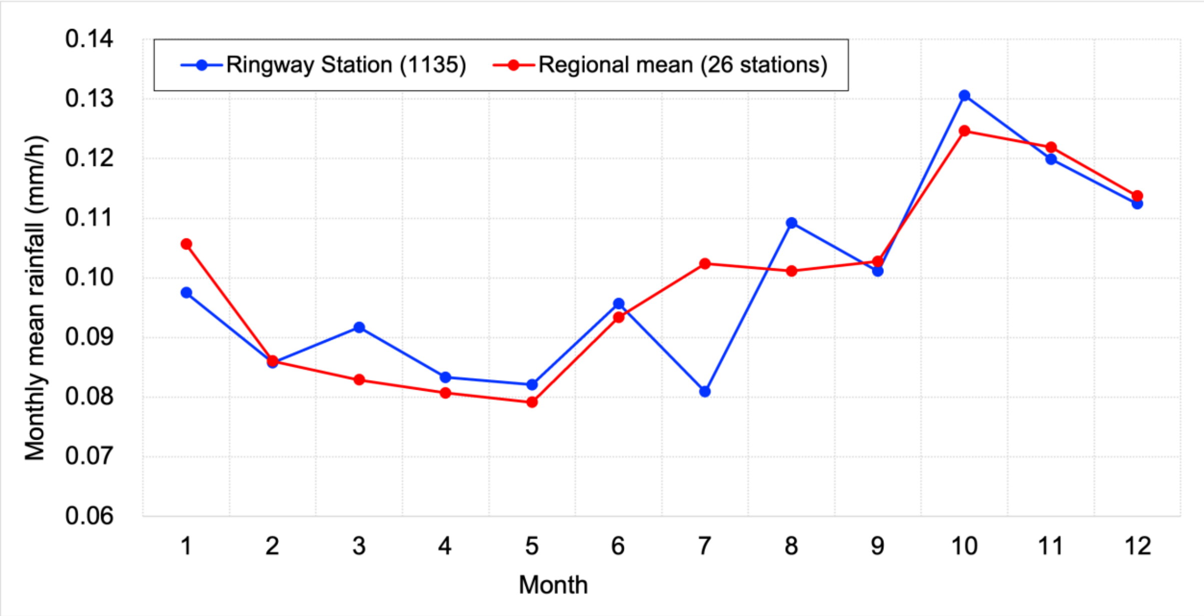


Figure X. Monthly mean rainfall at Ringway station (blue) and the regional mean of 26 surrounding stations (red) for 1981–2010. Both series show wetter conditions in winter and drier conditions in summer, indicating that Ringway is representative of the regional climatology.

4.2.2 Linear relationship between rainfall statistics and regression analysis

A linear regression was employed to quantify the relationships between rainfall statistics and large-scale climate variables. This approach was chosen for its clarity and interpretability, and because it has been widely applied in stochastic rainfall modelling, where approximate linear relationships between rainfall statistics and their climatic drivers are frequently observed (Onof, C. et al., 2000; Onof, Christian & Wang, 2020; Park, Onof & Kim, 2019). Previous studies have shown that aggregated rainfall statistics at the monthly scale can be reliably approximated by linear functions, making this method suitable for both robust inference and transparent parameterization of the Bartlett–Lewis model (Onof, Christian & Wang, 2020).

To enhance the reliability of the statistics, the regression analysis didn't rely on a single station but instead incorporated all valid sites within a 100 km radius of the reference location. This approach reduces local sampling errors and reflects the assumption that nearby sites share similar rainfall–climate relationships (Fowler et al., 2005). Moreover, separate regression analyses were conducted for each month, which allows the seasonal variations in the rainfall-climate ratio to be clearly demonstrated rather than being averaged over the annual cycle.

Two categories of regressions were considered. The first links mean rainfall to large-scale meteorological predictors derived from the UKCP18 CPM outputs, including near-surface temperature, relative humidity, wind speed, shortwave radiation, and longwave radiation. These variables were selected for their physical relevance: temperature and humidity constrain atmospheric moisture capacity and stability; wind speed modulates the advection and convergence of moist air; shortwave radiation governs surface heating and convective activity; and longwave radiation reflects cloud cover and radiative balance. By conducting a regression analysis between the average rainfall and these predictive factors, it is possible to capture how thermodynamic and dynamic processes regulate local precipitation.

The second regression type relates higher-order rainfall statistics, such as variance, skewness, and wet probability to mean rainfall itself. This formulation reflects well-documented scaling behaviours. For example, the positive association between mean rainfall and standard deviation indicates that wetter months tend to exhibit greater variability, consistent with the clustering of convective events (Park, Onof & Kim, 2019). Similarly, increasing skewness with higher mean rainfall suggests that wetter conditions are often accompanied by heavier-tailed distributions, indicative of more intense storms (Kim et al., 2017). The proportion of wet periods also rises with mean rainfall, linking total precipitation to storm frequency. In addition, cross-scale regressions between statistics at different aggregation durations (e.g., variance at 3, 6, or 12 hour intervals against the 1 hour variance) reveal how rainfall processes scale across temporal resolutions.

The regression model was expressed in a general form as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \varepsilon \tag{6}$$

where y denotes the dependent rainfall statistic (e.g., variance, skewness, or wet probability), x_i are independent predictors such as mean rainfall or meteorological variables, β_i are regression coefficients, and ε is the residual term. The residuals were subsequently modelled by resampling from their empirical distributions, generating multiple statistical ensembles to preserve stochastic variability around the fitted relationships (Onof, Christian & Wang, 2020). In this study, ten statistical ensembles were produced to reflect this uncertainty.

To illustrate the cross-scale regression analysis, Figures Y and Z present December as an example month. Five aggregation durations (1 h, 3 h, 6 h, 12 h, and 24 h) are shown for each statistic. Figure Y shows that the positive correlation between average rainfall and standard deviation is consistent across all scales, and the regression slope decreases as the aggregation increases. Figure Z shows the corresponding relationship with skewness. The skewness is weaker and more dispersed, but generally shows a negative trend. Other statistical data will be analyzed in the results section.

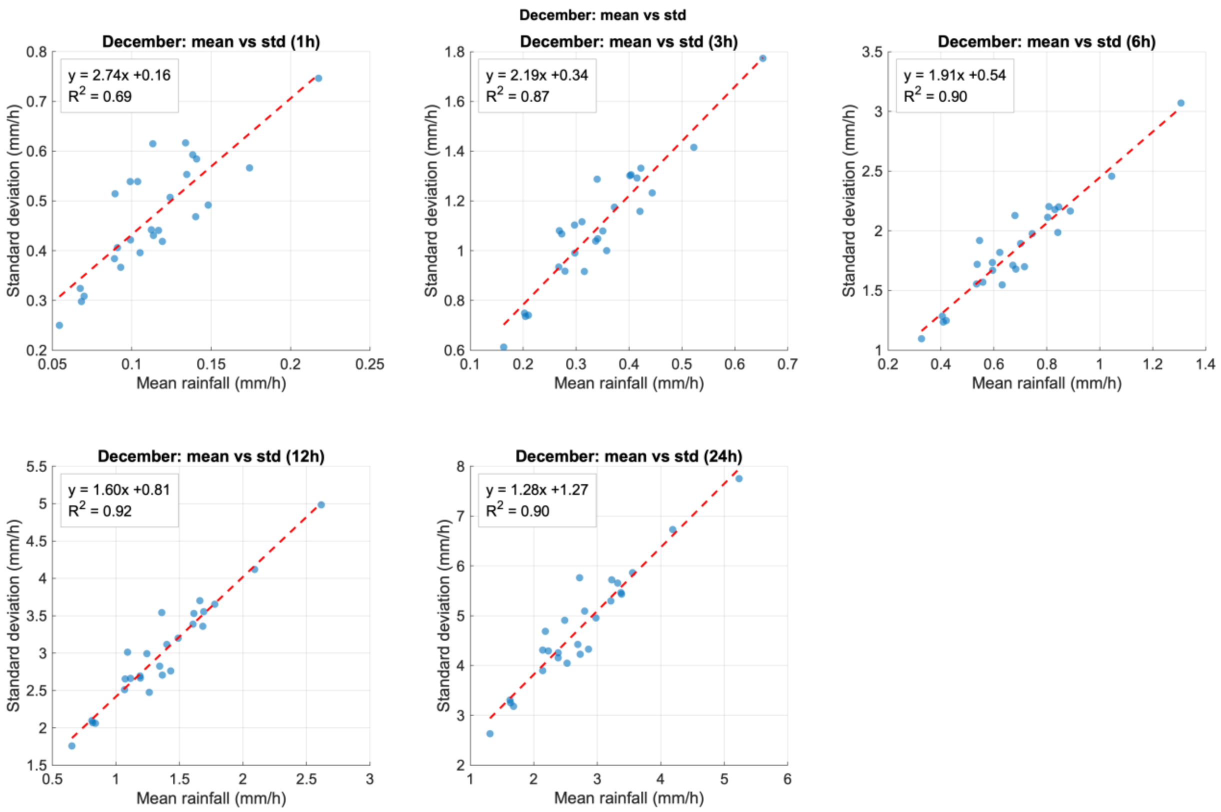


Figure Y Regression between mean rainfall and standard deviation for December, at five aggregation durations (1 h, 3 h, 6 h, 12 h, and 24 h). Regression equations and coefficients of determination (R^2) are shown in each panel. The results indicate that shorter durations exhibit stronger associations, while longer durations display lower slopes.

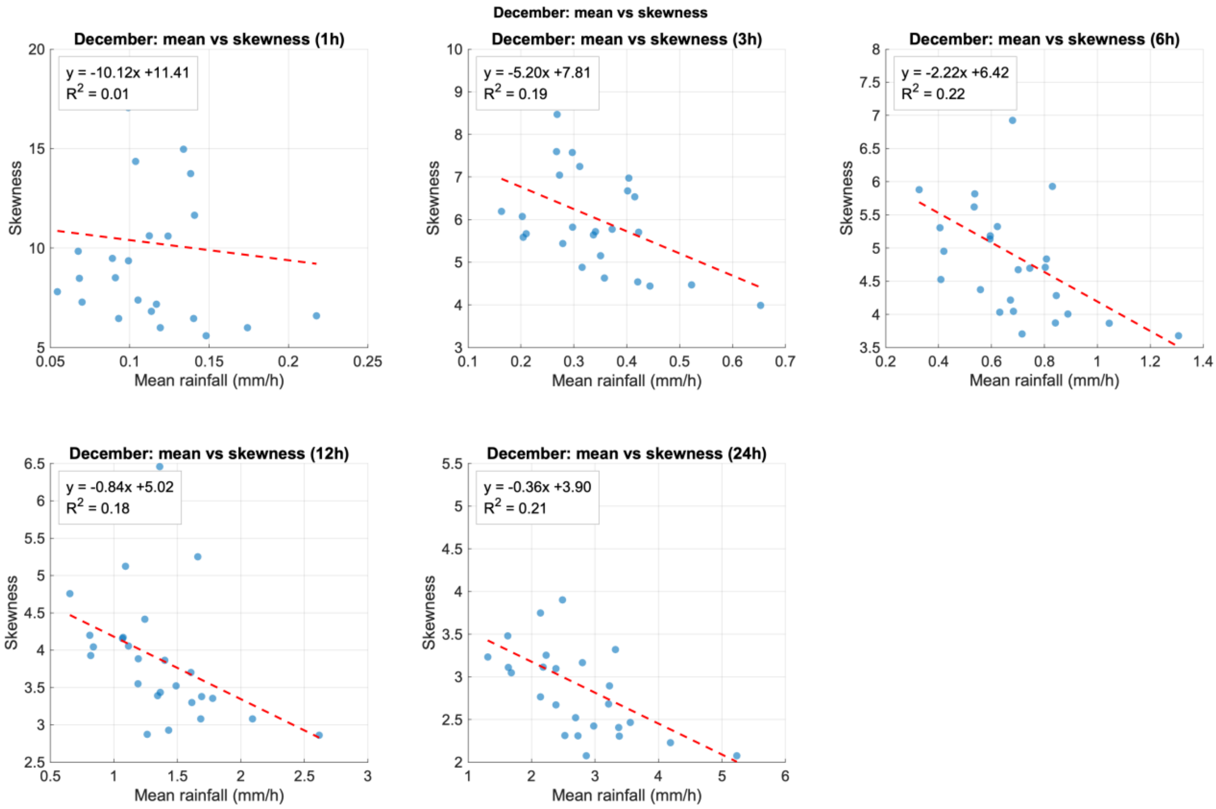


Figure Z Regression between mean rainfall and skewness for December, at five aggregation durations. The relationships are weaker and more scattered than for standard deviation, but generally show a negative association, especially at longer durations.

4.2.3 Projection of future rainfall statistics

Future rainfall statistics were derived using the change factor (CF) approach, which scales observed historical statistics according to the relative differences projected by climate models. For a given statistic S, the CF is defined in its simplest form as

$$CF = \frac{S_{future}}{S_{historical}} \tag{7}$$

where $S_{historical}$ and S_{future} represent the value of the statistic in the control period

(1981–2000) and future period (2061–2080) respectively, both derived from UKCP18 CPM outputs. The resulting CFs were then applied to the observed historical mean rainfall, while higher-order moments (variance, skewness, lag-1 autocorrelation, and wet probability) were reconstructed through the regression relationships described earlier. This ensures consistency across the full set of statistics used for stochastic model calibration.

The CF framework has been widely used in climate change impact studies because of several advantages. It is conceptually transparent and easy to apply, relying on simple ratios between historical and future climate statistics (Anandhi et al., 2011). It also retains the observed temporal and spatial structure of rainfall, making it compatible with stochastic weather generators that require consistency with baseline distributions (Wilby, Dawson & Barrow, 2002). Moreover, the method is computationally efficient, which is especially valuable when dealing with large climate model ensembles or multiple observational sites (Teutschbein & Seibert, 2012). Despite these strengths, this method assumes that the statistical relationships governing rainfall behavior are stable. This means that potential shifts in variability or extremes beyond those inferred from mean changes may not be adequately captured, leading to underestimation of climate change impacts in some contexts (Maraun et al., 2010).

The consistent findings in the study revealed a significant seasonal contrast for CF. The winter months typically showed stronger increases, reflecting the influence of strengthened westerly circulation and enhanced storm track activity. In contract, the summer months often showed weaker growth or even declines due to the prevalence of anticyclonic conditions and soil–atmosphere feedback (Hanlon et al., 2021; Kendon, Mike et al., 2023; Sexton et al., 2024). Observations also confirmed these trends, with UK winters becoming markedly wetter in recent decades (Carbon Brief). These results are consistent with broader European assessments, which indicate that compared to the summer, mid-latitude winter precipitation responds more strongly to climate warming (Christensen & Christensen, 2004; Frei et al., 2006).

4.3 Stochastic rainfall modelling

4.3.1 Model description

The Bartlett-Lewis clustering point process model family has been widely applied in generating random rainfall, providing a framework that balances physical interpretability and the ability to reproduce key rainfall statistics across different time scales. Within this family, the Rectangular Pulse formulation with randomised cell parameters (RBLRPx) represents an improved version that addresses limitations of earlier models, particularly in reproducing higher-order statistics such as skewness and lag-1 autocorrelation with fine temporal resolutions (Kim & Onof, 2020; Onof, Christian & Wheater, 1994; RODRIGUEZ-ITURBE, COX & ISHAM, 1987). For the purposes of this study, which requires long synthetic rainfall series for assessing changes in extremes under future climate scenarios. RBLRPx provides the necessary flexibility to capture both the mean behaviour and variability of hourly rainfall.

The RBLRPx model is constructed based on a storm-unit-pulse hierarchical structure. Storms arrive according to a Poisson process. Each storm initiates a cluster of rain cells, and within each cell, rectangular pulses of constant intensity are generated. The intensity and duration of these pulses are random variables, allowing the model to reproduce a wide range of rainfall patterns. Figure X illustrates the hierarchical nature of the model, which descripts how storm arrivals cascade into multiple cells and subsequently leads to rainfall pulses. This structure reflects the observed rainfall processes, where large-scale weather systems trigger convective events that produce bursts of rainfall at finer scales.

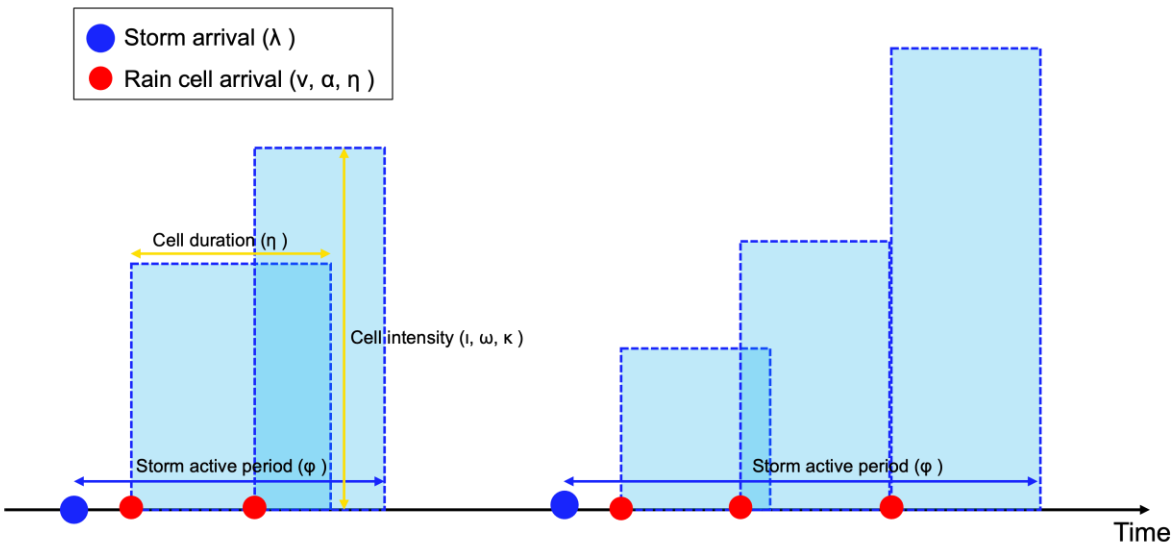


Fig. X Schematic representation of the RBLRPx model showing storm arrivals, rain cell arrivals, and rectangular pulses with variable duration (η) and intensity (ι, ω, κ).

Each parameter of RBLRPx controls different aspects of the rainfall process. The storm arrival rate (λ [1/T]) controls the frequency of storm clusters, while ν [–] is the Gamma shape parameter for the temporal scaling factor η, which governs the organization of cells within the storm. The parameter ι [L] defines the mean cell intensity coefficient (average rainfall magnitude), and ϕ [-] acts as a scaling factor for storm activity duration. The parameter κ [–] is a scaling factor for cell arrival rate, determining the clustering of pulses and influencing the storm duration. Two other parameters, α [1/T] (Gamma rate parameter for η) and ω [–] (Gamma shape parameter for rainfall cell intensity distribution), introduce randomisation into the intensity and duration of cells, improving the model's ability to reproduce skewness and variability observed in empirical data. The temporal scaling factor for cell duration, denoted by η, is modelled as a Gamma-distributed random variable with shape parameter ν and rate parameter α. Table Y summarizes these parameters, their definitions, and their roles in reproducing calibration targets (such as variance, skewness, lag-1 autocorrelation, and wet probability).

Parameters of the RBLRPx model: definitions, units, and physical interpretations.

Symbol	Unit	Definition	Physical meaning / Role
λ	[1/T]	Storm arrival rate	Controls the frequency of storm clusters

			(how often storms occur).
ν	[–]	Gamma shape parameter for η	Determines variability in the number of cells per storm.
α	[1/T]	Gamma rate parameter for η	Sets the time scale of cell arrivals, paired with ν.
ι	[L]	Mean cell intensity coefficient	Governs the average rainfall magnitude of individual cells.
φ	[–]	Scaling factor for storm activity duration	Controls the relative timing of cell activity within storms, shaping storm temporal structure.
κ	[–]	Scaling factor for cell arrival rate	Controls the rate of cell arrivals within storms, influencing clustering and storm duration.
ω	[–]	Gamma shape parameter for rainfall cell intensity distribution	Governs the skewness and extremes of rainfall by shaping the intensity distribution.

The choice of RBLRPx in this study is not only based on its previous successful record but also on its consistency with the modeling objectives. Simulating a 100-year hourly rainfall sequences requires a model with high computational efficiency, while still being able to capture extreme situations related to return period analysis. Simpler versions of the Bartlett–Lewis model, such as BLRP or MBLRP, have been shown to undersetimate high skewness or to fail to consistently capture multi-scale variance (Kaczmarska, Isham & Onof, 2014). In contrast, RBLRPx provides additional flexibility through random parameters, which is essential for reproducing the rainfall statistics. Moreover, its ability to generate physically interpretable processes makes it suitable for linking simulated rainfall with climate projections, ensuring transparency when applying change factors to future scenarios.

4.3.2 Calibration procedure

The calibration of the RBLRPx model was designed to ensure that the simulated rainfall time series can reproduce the key statistical characteristics of the observed or projected data. Calibration was performed separately for each month and for each ensemble member. This monthly and ensemble-based approach follows established practices in stochastic rainfall modelling (Cowpertwait, Paul SP, 1998; Onof, Christian

& Wheater, 1994), and captures both the seasonal variations in rainfall and the uncertainty arising from the climate projections.

The purpose of calibration is to minimize the difference between the simulated and target statistics. The five rainfall statistics indicators used as the target values not only reflect the central tendency and variability of rainfall, but also capture higher-order characteristics and temporal correlations. They ensure that the model can reproduce both average conditions as well as the persistence of wet and dry periods (Kaczmarska, Isham & Onof, 2014).

To balance the influence of different statistics, a standardized weighted least squares objective function was applied (Kim & Onof, 2020):

$$Obj(\theta) = \sum_{i} w_i \cdot \left(\frac{S_i^{sim}\ S_i^{target}}{S_i^{target}}\right)^2 \tag{8}$$

here S_i^{sim} is the simulated statistic, S_i^{target} is the observed or projected statistic, and w_i represents the weight assigned to each statistic. This formulation minimizes relative errors and prevents the calibration from being dominated by statistics with larger absolute values.

The optimization was carried out using the fminsearchbnd algorithm, which is a nonlinear search method with boundary constraints (Kim & Onof, 2020). It provides a practical balance between efficiency and robustness, particularly when the parameter space is large and the objective surface is complex.

The result of this process is a set of calibration parameter vectors for each month and ensemble. The calibration performance can be evaluated by comparing the observed and simulated statistics or by visualizing the convergence of the parameter trajectories during the optimization process. This provides confidence for the next step of generating long synthetic rainfall sequences.

4.3.3 Simulation of synthetic rainfall series

Once the parameters of the RBLRPx model were calibrated, synthetic rainfall series were generated, reproducing the observed statistics at hourly resolution. Storm arrivals were modelled as a Poisson process, where each storm initiated a cluster of rain cells and subsequent rectangular pulses of intensity and duration (Onof, Christian & Wheater, 1994; RODRIGUEZ-ITURBE, COX & ISHAM, 1987). Each parameter governed specific aspects of this process, as described previously, and together they determined the temporal evolution of simulated rainfall events.

Using the corresponding calibration parameter set, simulations are conducted independently for each calendar month. This ensured that seasonal variations of rainfall characteristics were preserved in the generated sequences. Then, the monthly simulations were connected together to construct continuous series spanning 100 years, providing a powerful dataset for statistical and extreme value analysis.

To reflect the uncertainty of the parameter, multiple stochastic realizations were generated by repeating the simulation process with different random seeds, following practices adopted in previous stochastic rainfall studies (Cowpertwait, Paul SP, 1998; Kaczmarska, Isham & Onof, 2014; Kim & Onof, 2020). This method produced a series of rainfall sequences, which can quantify the variability in the simulation statistics and enhance the reliability of subsequent climate impact assessments.

4.3.4 Post-processing and analysis

After generating the synthetic rainfall series, several post-processing steps were carried out to evaluate model performance and obtain information relevant for subsequent analyses. Firstly, the simulated series were subject to statistical verification to assess whether the target statistics used for calibration were adequately preserved. This involved comparing the simulated and observed (or projected) means, variances, skewness, lag-1 autocorrelations, and wet probabilities across different temporal scales. Secondly, extreme value analysis was performed on the simulated series in order to estimate indicators related to the design, such as return levels associated with 10-year, 20-year, or 50-year rainfall events. Thirdly, the historical and future simulations were compared in terms of changes in their statistical characteristics, using tables and graphs to highlight seasonal and annual differences. Finally, the broader implications of these findings were considered in the context of regional water resource management and design adaptation, linking the changes in rainfall characteristics to flood risk assessment, infrastructure resilience, and practical challenges in planning.